CS341 · COMPUTER GRAPHICS

Midterm 462 Solutions

20 marks · 1 hour

Muharib

01 Solution Q1: fill in the gaps

B-Reps, voxels, CSG and BSP are not in this term's slides, so their answers follow the textbook. They are outside the midterm scope.

1. ... piecewise smooth surfaces like **polygons (triangle meshes)** or **spline (Bézier) patches**.

2. Scan conversion converts **geometric (mathematical / continuous)** representations of shapes into **pixels** on a raster display.

3. The next pixel is either **(x+1, y)**, horizontal (same row), or **(x+1, y+1)**, diagonal (one row up).

02 Solution Q2-A: true / false

#	Answer	Reason
1	**False**	Bresenham uses only integer calculations; DDA is the one that uses floating point.
2	**True**	A raster is a matrix of pixels (rows and columns).
3	**True**	Super-sampling takes several samples per pixel and averages them, reducing jaggies.
4	**False**	A voxel model is a volumetric (solid) representation, not a surface one.
5	**True**	A triangle is always planar and simple, and graphics hardware is built for it.
6	**False**	There are also CAD, medical visualization, simulation, education and more.

03 Solution Q2-B: multiple choice

#	Answer	Reason
1	**c) Scaling**	Translation moves, rotation turns, and blurring is not a geometric transformation.
2	**c) Voxel**	Voxels represent the inside of a volume, as in CT scans.
3	**b) The next pixel**	The sign of `p(k)` picks the next pixel.
4	**a)**	$F(x, y) = x^2 + y^2 - r^2$
5	**c) Mirroring**	A negative factor flips the shape (reflection).
6	**b)**	CSG combines simple shapes with union, intersection and difference.
7	**a)**	BSP recursively divides space into half-spaces using planes.

04 Solution Q3: short answers

A. Two advantages of triangle meshes

1. Simple and always planar (three points define one plane), so they are easy to process and render.
2. Supported and highly optimized by graphics hardware (GPUs).
3. Any surface can be approximated by enough triangles.

B. The two main 3D representations

Boundary representation (B-Reps): describes only the surface (boundary) of the object, e.g. polygon / triangle meshes.
Solid (volumetric) representation: describes the interior volume of the object, e.g. voxels, CSG, BSP trees, octrees.

C. Two applications of scan conversion

Drawing lines and circles on a raster screen (DDA, Bresenham); filling polygons; rendering text (fonts); rasterizing 3D scenes for display.

05 Solution Q4-A: rotation about a point

Rotation about a point other than the origin takes three steps: translate (1,1) to the origin, rotate, translate back.

```
M = T(1, 1) · R(90°) · T(−1, −1)
```

```
R(θ):  x' = x cos θ − y sin θ ,   y' = x sin θ + y cos θ
```

```
cos 90° = 0 ,  sin 90° = 1
```

```
1) T(−1, −1):   (5, 3) → (5 − 1, 3 − 1) = (4, 2)
```

```
2) R(90°):      x' = 4(0) − 2(1) = −2
```

```
                y' = 4(1) + 2(0) = 4      → (−2, 4)
```

```
3) T(1, 1):     (−2 + 1, 4 + 1) = (−1, 5)
```

With the composite homogeneous matrix:

$$M = \begin{vmatrix} 0 & -1 & 2 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} \cdot \begin{vmatrix} 5 \\ 3 \\ 1 \end{vmatrix} = \begin{vmatrix} 0\cdot5 - 1\cdot3 + 2 \\ 1\cdot5 + 0\cdot3 + 0 \\ 1 \end{vmatrix} = \begin{vmatrix} -1 \\ 5 \\ 1 \end{vmatrix}$$

$$P' = (-1, 5)$$

06 Solution Q4-B: Bresenham

$$\Delta x = 10 - 3 = 7 \qquad \Delta y = 7 - 3 = 4 \qquad m = \frac{4}{7} < 1$$

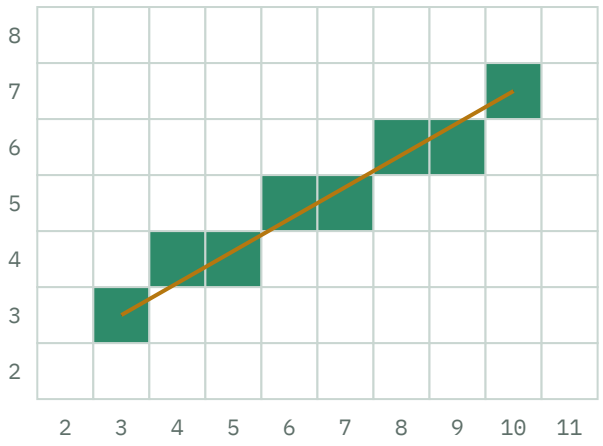
$$2\Delta y = 8 \qquad 2\Delta y - 2\Delta x = -6$$

$$p0 = 2\Delta y - \Delta x = 8 - 7 = 1$$

$$p < 0 \rightarrow (x+1, y) \qquad p = p + 8$$

$$p \geq 0 \rightarrow (x+1, y+1) \qquad p = p - 6$$

k	p_k	Decision	Next pixel	p_{k+1}
0	1	$p_k \geq 0$	(4, 4)	-5
1	-5	$p_k < 0$	(5, 4)	3
2	3	$p_k \geq 0$	(6, 5)	-3
3	-3	$p_k < 0$	(7, 5)	5
4	5	$p_k \geq 0$	(8, 6)	-1
5	-1	$p_k < 0$	(9, 6)	7
6	7	$p_k \geq 0$	(10, 7)	1




```
Pixels: (3,3) (4,4) (5,4) (6,5) (7,5) (8,6) (9,6) (10,7)
```